BSBS • SYSTEM DEVELOPMENT PROJECT

P2P RIDE-SHARING FOR BOTSWANA

A geolocation-based mobile app that matches drivers' spare seats with passengers on compatible routes to share commuting costs

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TECH STACK

Expo + React Native + Supabase + Google Maps API

DURATION

3 months

SUPERVISOR

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1 INTRODUCTION & PROBLEM STATEMENT

Background

- Commuters in Gaborone depend on private vehicles or public transport, which can be **costly and hard to coordinate**.
- Ride-sharing lets people on similar routes **share vehicle capacity**, cutting costs and vehicles needed (Agatz et al., 2012).

Problem Statement

- Botswana lacks a **convenient, localised and affordable** peer-to-peer ride-sharing service for everyday commuters.
- Ride-hailing apps dispatch a driver to a passenger; they do not match **seats on journeys already planned** with compatible passengers.

2 LITERATURE REVIEW

- **Dynamic ride-sharing** pairs travellers with similar itineraries and schedules (Agatz et al., 2012).
- Good matches need **route, time, seat and detour** compatibility, not proximity alone (Hong et al., 2017; Stiglic et al., 2016).
- **Gap:** no local platform built on existing journeys and available seats.

System	Strength	Limitation
Uber / Bolt	On-demand requests, live driver matching, scheduled rides	Driver travels to the passenger; not built on seats on planned journeys
inDrive	Passenger-proposed prices; drivers accept, decline or counter	Price negotiation rather than route and seat compatibility
BlaBlaCar	Drivers publish journeys with seats; costs are shared	Full platform; not tailored to urban commuting in Botswana
Proposed artifact	Matches drivers' planned journeys with passengers by route, time and seats for local commuting	

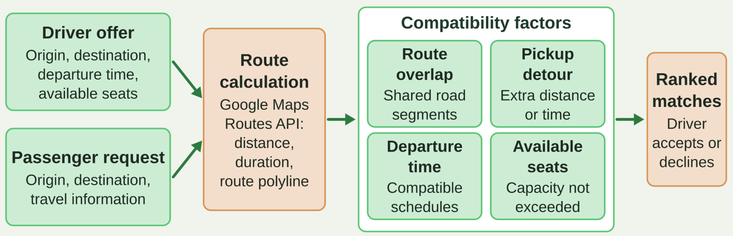


Figure 1. [Conceptual framework]

3 SYSTEM DESIGN

- **16 functional** and **8 non-functional** requirements.
- Drivers offer seats; passengers request rides.
- Matching on route, time, seats and detour.
- **Expo + React Native** app for Android and iOS.
- **Supabase** for authentication and PostgreSQL data.
- **Google Maps Routes API** for routes.



Figure 2. [High-level architecture]

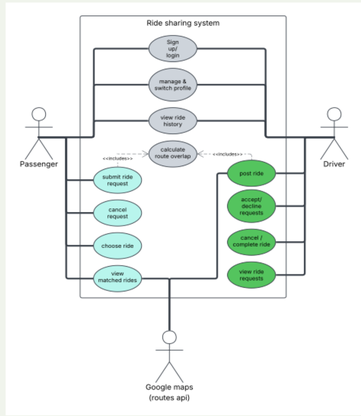


Figure 3. [Use case diagram]

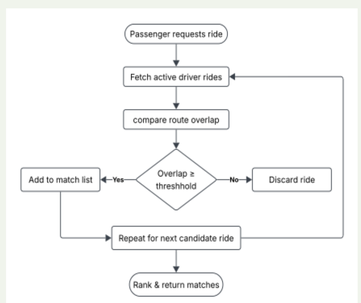


Figure 4. [System flowchart]

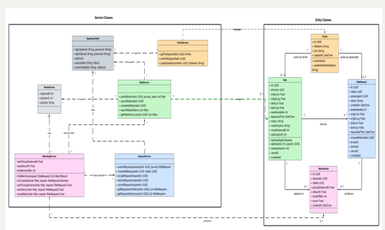


Figure 5. [Class diagram]

4 RESULTS & DISCUSSION

- **Status:** requirements and literature review documented; prototype under development.
- **Evaluation plan:** test cases, a **SUS usability survey** and response-time testing.
- Whether commuters find the approach useful will be judged through **testing and user feedback**.

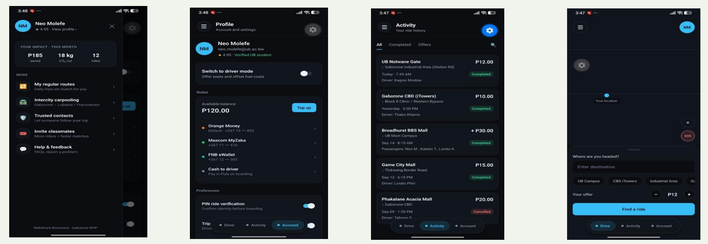


Figure 6. [The working artifact in use]

Criterion	Method	Target	Result
Functionality	Test cases (RQ01–RQ16)	All core requirements met	Core features working; matching in progress
Usability	SUS survey and core-task testing	Core tasks completed with minimal difficulty	Pending user testing
Performance	Match response time (NFR01)	Within the defined prototype target	Matching slow; being optimised

5 CONCLUSION

- Effective ride-sharing needs **journey compatibility**, not proximity alone.
- Contribution: a **localised P2P commuting app** built on existing journeys.
- Requirements are defined; testing will show whether they are met.

6 RECOMMENDATIONS & FUTURE WORK

- **Recommend** simple, measurable and explainable matching rules.
- **Limits:** relies on connectivity, the Routes API and GPS accuracy.
- **Next: live ride tracking** (like Uber and inDrive) and driver and passenger rating.

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https://github.com/K-Serumola/KJJB